

LabVIEW-Controlled Smart Garden Robot

with Self-Built Soil Moisture Sensor, AI Weed Detection, and Automatic Irrigation Control

Full Mini Project Proposal
Electrical Measurements and Instrumentation

Sensor Signal Conditioning DAQ/Arduino LabVIEW Decision Logic Actuator

Prepared for

University of Moratuwa, Sri Lanka
B.Sc. Engineering Degree Course - Semester 4
EE2043 / EE2044 Electrical Measurements and Instrumentation

Project focus

- Autonomous pot plant monitoring using four sensors and multiple actuators.
- Instrumentation contribution: self-built calibrated soil moisture sensor with conditioning.
- Control contribution: automatic pump duration calculation and LabVIEW data logging.
- Software contribution: camera-based weed detection and local web dashboard.

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Executive Summary and Contents

A complete proposal aligned with the mini project brief

Project title: LabVIEW-Controlled Smart Garden Robot with Self-Built Soil Moisture Sensor, AI Weed Detection, and Automatic Irrigation Control.

This proposal presents a compact smart garden instrumentation system for monitoring pot plants. The robot stops at each pot station, measures soil moisture and temperature, captures an image for weed detection, updates a local website, logs data through LabVIEW, and automatically controls a water pump according to the measured soil moisture percentage.

The main technical contribution is not only robot movement or AI. The system is framed as a complete measurement and control chain: sensor, signal conditioning, DAQ/Arduino acquisition, LabVIEW analysis, decision logic, and actuator response. This directly supports the instrumentation requirement of the mini project.

Proposal compliance snapshot

- 3+ sensors included
- Arduino Uno control
- NI DAQ measurement
- Performance testing plan
- 2+ actuators included
- LabVIEW programming
- Self-built sensor + conditioning
- Classroom demo workflow

Contents

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Project Background and Problem Statement

Why the system is needed and what problem it solves

Small potted plants in homes, nurseries, schools and research gardens require regular monitoring. In a manual system, each plant must be inspected for soil dryness, ambient temperature and weed growth. This is repetitive, time-consuming and prone to delayed action. When soil becomes too dry, the plant experiences water stress; when weeds grow near the plant, they compete for water, nutrients, light and space.

The proposed solution is a smart garden robot that moves between pot stations and performs measurements at each plant. It captures a plant image, classifies the presence of weeds, measures soil moisture and temperature, displays calibrated values in LabVIEW, and activates a small water pump when the soil moisture is below a selected target. The system also updates a local web dashboard to show weed alerts and plant condition.

Problem statement

Manual pot plant monitoring is inefficient because soil moisture, temperature condition and weed growth must be checked repeatedly. A LabVIEW-controlled instrumentation robot can automate sensing, decision-making, irrigation and status reporting for multiple pot plants.

The proposed work is suitable for a mini project because it combines measurable physical quantities, sensor calibration, signal conditioning, software-based acquisition, actuator control and performance evaluation. The design remains achievable by limiting the demonstration to 3-5 pot plants placed in a controlled indoor line or station arrangement.

The system should be presented as an instrumentation system rather than only a robot or only an AI project. The key project chain is shown below.



Core instrumentation chain used to present the project to the lecturer

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Aim, Objectives, Scope and Brief Compliance

Clear deliverables for the project and module requirements

Aim: To design, build and test a LabVIEW-controlled smart garden robot that monitors pot plants using multiple sensors, detects weeds using image analysis, and automatically irrigates plants according to calibrated soil moisture readings.

Specific objectives

- Design and build a self-made soil moisture sensor using stainless-steel probes, signal conditioning and calibration.
- Measure soil moisture percentage and environmental temperature using DAQ/LabVIEW-based acquisition.
- Use a camera-based method to classify plant images as weed detected, no weed or invalid image.
- Control robot movement so the system stops reliably at each pot station.
- Calculate pump activation duration based on moisture deficit and prevent overwatering using a maximum time limit.
- Display sensor values, control status, graphs and logs on a LabVIEW front panel.
- Update a local website dashboard with pot ID, weed status, moisture percentage, temperature and watering action.
- Evaluate the system using sensor, control, AI and robot performance tests.

Scope of the prototype

Included in scope	Not included in first prototype
3-5 pot plants arranged in a line or marked stations.	Outdoor full-garden navigation or rough terrain operation.
Self-built resistive soil moisture sensor with conditioning.	Commercial soil sensor as the main contribution.
LM35 or DHT temperature measurement.	Complete greenhouse climate control with heating/cooling.
Camera-based weed/no-weed detection.	Complex robotic weed removal or cutting mechanism.
LabVIEW dashboard, DAQ reading and data logging.	Cloud deployment, mobile app or remote IoT database.
Pump duration control based on moisture deficit.	Closed-loop high-precision irrigation with flow meter feedback.

Compliance with the mini project brief

Brief requirement	How this proposal satisfies it
At least 3 sensors	Self-built soil moisture sensor, temperature sensor, camera, and IR/ultrasonic pot marker sensor.
At least 2 actuators	Robot drive motors and water pump. Optional buzzer, LED and servo may be added.
Arduino Uno control	Arduino controls motors, pump driver, marker sensor and serial communication.
LabVIEW programming	LabVIEW displays measurements, performs logic, logs data and sends commands.
NI DAQ card use	DAQ reads analog sensor outputs such as soil moisture voltage and LM35 voltage.
Self-built sensor recommended	A soil moisture sensor is built from stainless-steel probes with voltage divider, LM358 buffer and RC filter.
Workflow diagram and simulator testing	Arduino control sequence is prepared and can be verified in LabVIEW Arduino Simulator before hardware integration.
Assess performance parameters	Calibration, repeatability, response time, pump timing, AI accuracy and stopping accuracy are tested.

Why this proposal is defensible

The project directly shows measurement, calibration, signal conditioning, DAQ acquisition, LabVIEW decision logic and actuator response. This makes it stronger than a simple robot demo because every major action is linked to a measured sensor value.

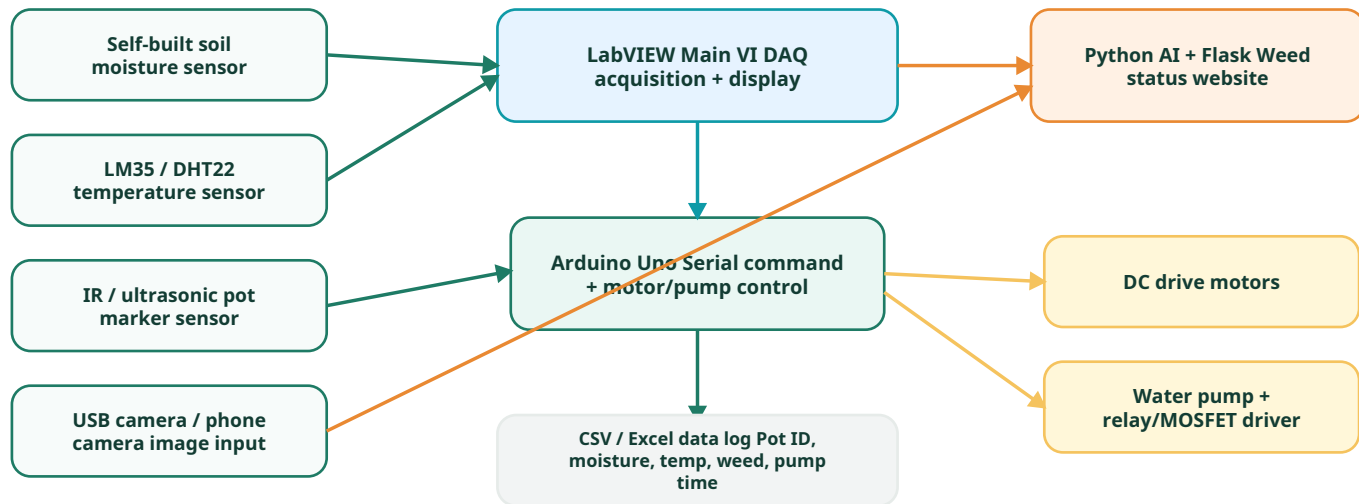
3

Proposed System Concept and Architecture

How sensors, software and actuators work together

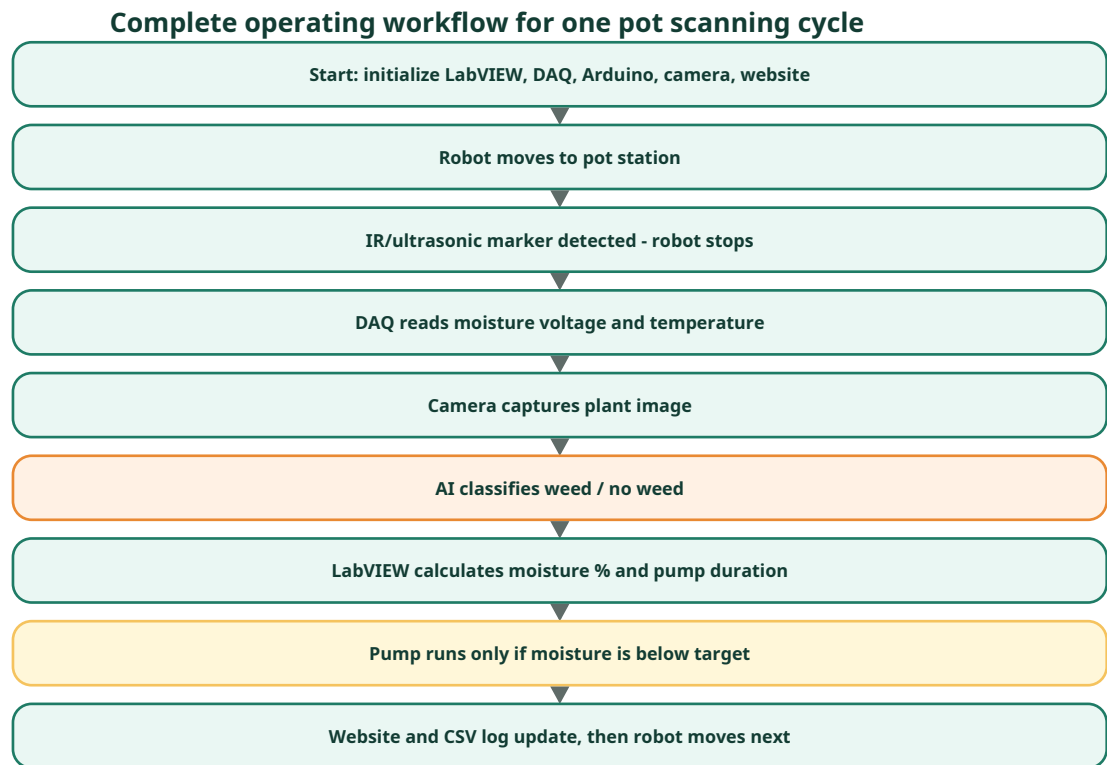
The robot follows a station-based navigation approach. Each pot has a simple marker, such as black tape or a fixed distance point. When the IR or ultrasonic sensor detects a pot station, the robot stops and the measurement cycle begins. LabVIEW acquires soil moisture and temperature data, Python analyzes the image for weed status, and Arduino controls the pump and drive motors.

Proposed system architecture



The architecture separates the system into three practical layers. The **measurement layer** contains soil moisture, temperature, camera and navigation sensors. The **processing layer** contains LabVIEW, NI DAQ, Arduino serial communication and Python image analysis. The **actuation and reporting layer** contains robot motion, pump operation, LED/buzzer indication, data logging and the web dashboard.

Final system workflow



This controlled workflow reduces demonstration risk because the robot does not need complex mapping. It only needs to move from one marked pot station to the next, stop, measure, decide, water if required, log the result and continue.

4

Sensor Design and Signal Conditioning

Measurement principles, calibration and equations

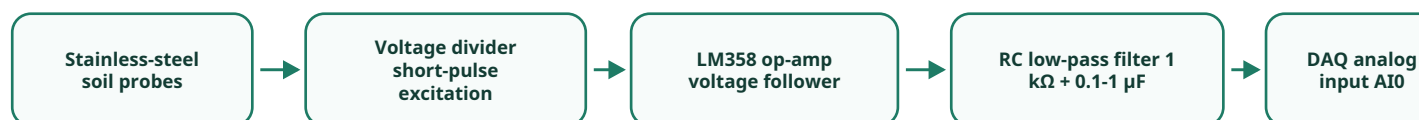
The sensor subsystem is the strongest instrumentation part of the project. Four sensors are recommended so that the minimum requirement is exceeded while keeping the prototype manageable.

Sensor	Type	Purpose	Interface
Self-built soil moisture sensor	Built by group	Measures soil moisture percentage using resistance change	Voltage divider + LM358 + RC filter to DAQ AI0
Temperature sensor	LM35 preferred / DHT22 alternative	Measures plant environment temperature	LM35 analog voltage to DAQ AI1 or DHT digital to Arduino
Camera / webcam	Existing image sensor	Captures plant image for weed detection	USB/phone camera to Python
IR or ultrasonic marker sensor	Existing navigation sensor	Detects pot station and stops robot	Arduino digital/analog input

4.1 Self-built soil moisture sensor

The proposed soil moisture sensor uses two stainless-steel probes inserted into the soil. Soil moisture affects the electrical resistance between the probes. Dry soil has higher resistance, while wet soil has lower resistance. By placing the soil resistance in a voltage divider, the output voltage changes with moisture level.

Self-built soil moisture sensor and conditioning path



Reason for conditioning: stable sensor voltage, lower noise, safer DAQ input, and demonstrable instrumentation content.

To reduce electrode corrosion, the sensor should not be powered continuously. For the prototype, the Arduino can energize the divider only during measurement for approximately 1-2 seconds, then switch it off again. This is safer for the electrodes and demonstrates good practical instrumentation thinking.

Voltage divider relation

$$V_{out} = V_{in} \times \frac{R_{fixed}}{R_{soil} + R_{fixed}}$$

R_{soil} is the resistance between probes, R_{fixed} is a known resistor such as 10 kΩ or 47 kΩ, V_{in} is normally 5 V, and V_{out} is read by the DAQ/Arduino. The exact direction of voltage change depends on the divider arrangement.

Signal conditioning: The divider output is connected to an LM358 voltage follower to avoid loading the sensor. A simple RC low-pass filter, such as 1 kΩ with 0.1-1 μF, reduces noise and gives a stable analog reading for the DAQ input.

4

Sensor Calibration and Temperature Measurement

Converting raw voltage into useful engineering values

4.2 Soil moisture calibration

Calibration is essential because the self-built moisture sensor output depends on soil type, probe spacing, electrode material and water content. The project should prepare four test samples: dry soil, low moisture soil, medium moisture soil and wet soil. Each condition is measured several times and an average voltage is recorded.

Condition	Preparation	Assigned value for relative calibration
Dry soil	Completely dry soil sample	0%
Low moisture	Add a small amount of water and mix uniformly	30%
Normal moisture	Add moderate water until suitable plant moisture	60%
Wet soil	Fully watered soil without flooding the pot	100%

Gravimetric reference method

$$\text{Moisture \%} = \frac{W_{\text{wet}} - W_{\text{dry}}}{W_{\text{dry}}} \times 100$$

This method compares wet and dry soil mass. It can be used if the group has enough time to dry samples and create a stronger calibration reference.

Relative voltage calibration for mini project demo

$$\text{Moisture \%} = \frac{V_{\text{sample}} - V_{\text{dry}}}{V_{\text{wet}} - V_{\text{dry}}} \times 100$$

For a short prototype, dry and wet voltage endpoints can be used to map V_{sample} into 0-100%. The result must be limited to 0 Moisture% 100.

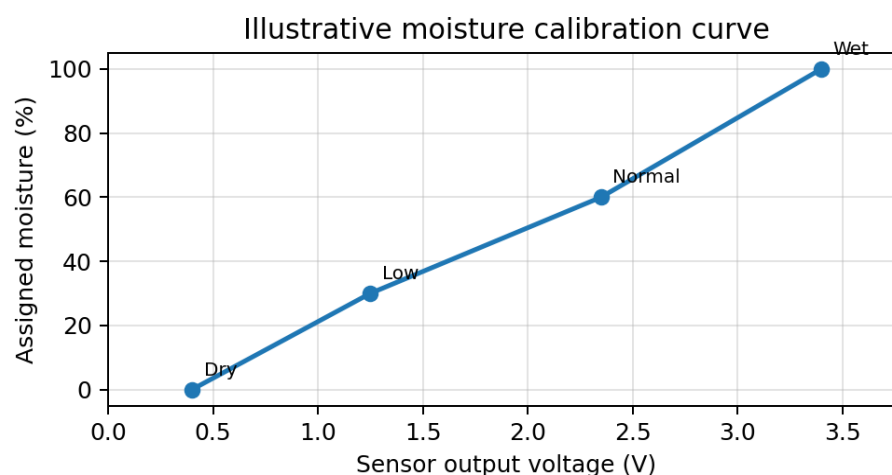


Figure: expected calibration curve style. The final report should replace these illustrative values with measured calibration data.

4.3 Temperature measurement

The LM35 is preferred for this instrumentation project because it provides an analog voltage proportional to temperature, allowing the DAQ card to acquire it directly. A DHT11/DHT22 can also be used if humidity is desired, but the LM35 better supports analog measurement practice.

LM35 conversion equation

$$T(^{\circ}\text{C}) = \frac{V_{out}(\text{mV})}{10}$$

Example: if $V_{out} = 300 \text{ mV}$, then $T = 30^{\circ}\text{C}$. The reading can be compared with a thermometer to estimate accuracy.

Temperature sensor verification

Check	How to perform it	Expected evidence
Room temperature reading	Place LM35/DHT near the pot and record stable value	LabVIEW displays a realistic indoor temperature
Thermometer comparison	Compare sensor reading with a simple thermometer	Report approximate error in $^{\circ}\text{C}$
Response observation	Warm the sensor gently using hand warmth, not flame	Temperature graph changes smoothly
DAQ scaling	Convert voltage to temperature inside LabVIEW	No manual calculation during demo

5

Actuator and Control Design

Robot movement, irrigation control and safe driver circuits

The project uses at least two actuators: DC drive motors for robot movement and a mini water pump for automatic irrigation. Additional actuators such as buzzer, status LEDs or a servo can be added if time remains.

Actuator	Purpose	Control method
DC drive motors	Move robot from one pot station to the next	Arduino + L298N motor driver
Mini water pump	Water plant when moisture is below target	Arduino digital output + MOSFET/relay driver
Buzzer	Alarm for weed or low moisture	Arduino digital output
LED indicators	System state, warning and pump activity	Arduino/DAQ digital output
Servo motor (optional)	Camera angle control or weed marker pointer	Arduino PWM

5.1 Automatic irrigation logic

Instead of simply switching the pump on or off, the pump duration is calculated based on how far the measured moisture is below the target value. This makes the system more intelligent and prevents unnecessary watering.

Moisture deficit and pump duration

$$D = M_{target} - M_{measured}, \quad t_{pump} = \min(KD, t_{max})$$

Recommended starting values: $M_{target} = 60\%$, $K = 0.12 \text{ s}/\%$ and $t_{max} = 8 \text{ s}$. If $D \leq 0$, the pump remains OFF.

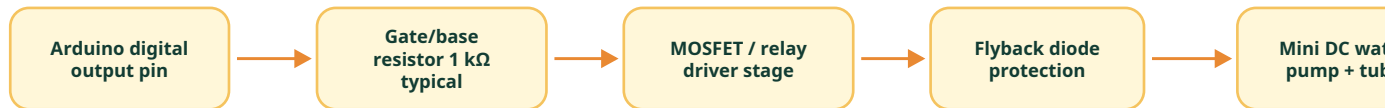
Measured moisture %	Condition	Control action
0-20%	Very dry	Pump ON up to 8 s
21-40%	Dry	Pump ON about 4-5 s
41-60%	Slightly low	Pump ON about 1-3 s
>60%	Normal	Pump OFF

Worked example: If $M_{target} = 60\%$ and $M_{measured} = 30\%$, then $D = 30\%$. With $K = 0.12 \text{ s}/\%$, the pump duration becomes 3.6 s, rounded to approximately 4 s. If the calculated duration exceeds 8 s, it is limited to 8 s.

5.2 Pump driver safety

The water pump must not be connected directly to the Arduino or DAQ output. A driver circuit is required because the pump draws more current than a microcontroller pin can safely supply. A MOSFET or relay module is recommended, with a separate pump supply and common ground.

Safe pump actuator driver concept



The pump is not powered directly from the Arduino or DAQ. A separate pump supply and common ground are used.

Important wiring rule

Arduino ground, pump supply ground and DAQ reference must be handled carefully. The pump should use an external supply suitable for its voltage rating, and a flyback diode should be used across inductive loads to protect the circuit.

5.3 Robot movement strategy

Navigation should be kept simple and reliable. The recommended method is a line of pot stations with black tape markers or fixed distance points. The robot moves slowly, detects the marker, stops, performs the sensing cycle and then moves to the next marker.

Movement stage	Arduino action	LabVIEW/DAQ action
Move forward	Drive motors through L298N	Display robot state as MOVING
Marker detected	Stop motors and send POT_REACHED	Start sensor acquisition and image capture
Measurement complete	Wait for command	Calculate pump duration and log sensor values
Watering complete	Send PUMP_DONE	Update pump status and command NEXT_POT

6

LabVIEW, DAQ, Arduino, AI and Website Integration

Software roles and data flow

6.1 LabVIEW front panel role

LabVIEW should be the main instrumentation dashboard. It should not be limited to a simple serial monitor. The front panel should clearly display raw voltage, calibrated moisture percentage, temperature, weed result, pump status, robot state, graphs, manual controls and data logging options.

Front panel section	Recommended indicators and controls
Sensor readings	Soil moisture voltage, calibrated moisture %, temperature in °C
AI status	Image captured, weed detected/no weed, confidence score, pot number
Actuator control	Pump ON/OFF, pump duration, motor movement, buzzer/LED state
Graphs	Moisture vs time, temperature vs time, pump duration per pot
Manual override	Stop robot, move next pot, pump ON/OFF, capture image again
Data logging	Save CSV/Excel file with timestamp, pot ID and action

6.2 NI DAQ card usage

DAQ channel	Connection	Measurement/control purpose
AI0	Conditioned soil moisture sensor output	Moisture voltage and calibrated moisture %
AI1	LM35 temperature sensor output	Environmental temperature
AI2 optional	LDR or additional analog sensor	Light availability or extra plant health variable
DO0 optional	Pump relay/MOSFET command	Pump control if DAQ digital output is used
DO1 optional	LED/buzzer command	Alarm or status indication

6.3 Arduino role

Arduino handles real-time low-level control tasks, including robot motors, pump driver output, marker sensor reading and serial communication with LabVIEW. LabVIEW sends commands such as MOVE, STOP, PUMP_ON_4, NEXT_POT and BUZZER_ON. Arduino returns messages such as POT_1_REACHED, PUMP_DONE, OBSTACLE_DETECTED and BATTERY_LOW.

6.4 AI weed detection approach

For the mini project, a reliable lightweight approach is better than an advanced but unstable model. Two options are proposed. Option 1 is image classification using three classes: plant only, plant with weed and soil/background. Option 2 is an OpenCV crop-zone method that detects extra green regions outside the main plant area.

Method	How it works	Best use
Simple image classifier	Train using plant only, plant + weed and soil/background images	Good if the group can collect 80-130 images
OpenCV crop-zone method	Segment green color, identify main plant region, detect extra green areas near soil	Good for fast, controlled pot demo with same camera angle
Confidence score	Convert model output into no weed / possible weed / weed detected	Improves dashboard and presentation quality

6

Website Dashboard and Data Logging

How plant status is stored and displayed

A local Flask website is sufficient for the prototype. Python receives or generates the weed detection result and writes plant status to a CSV/JSON file. The website reads this file and displays the latest plant condition in a simple dashboard.

Website field	Example display
Pot ID	Pot 1, Pot 2, Pot 3
Weed status	No weed, Possible weed, Weed detected
Soil moisture	35%, 67%, 52%
Temperature	30.2°C, 29.5°C, 31.0°C
Watering status	Pump ON 4 s or No watering
Alert section	Weed detected in Pot 2; Low moisture in Pot 3

Recommended CSV logging format

Time	Pot ID	Moisture %	Temp °C	Weed status	Pump duration	Action
10:01	Pot 1	35	30.2	Weed	3.0 s	Pump ON
10:05	Pot 2	68	29.5	No weed	0 s	No action
10:09	Pot 3	42	31.0	Weed	2.2 s	Pump ON

The logged data are useful for the final presentation because they show that the system performed actual measurements and decisions, not only manual demonstrations. Graphs can be generated from these values in LabVIEW or Excel.

LabVIEW block diagram logic

- Initialize DAQ channels, serial port, camera process and data log file.
- Read soil moisture voltage and temperature voltage continuously or after robot stop.
- Convert raw voltages into calibrated moisture percentage and temperature.
- Receive pot station status from Arduino and weed result from Python/CSV.
- Calculate moisture deficit and pump duration.
- Send pump and movement commands to Arduino through serial communication.
- Update indicators, graphs, alarms and CSV/Excel log.
- Repeat until all pots are scanned or the stop button is pressed.

Presentation tip

During the demo, keep LabVIEW open as the main screen. Show the live sensor voltage, calculated moisture percentage, temperature, weed status, pump command and data log changing for each pot.

7

Methodology, Testing and Performance Evaluation

How the system will be built and proven

7.1 Development methodology

Stage	Main tasks	Deliverable
Design	Finalize pot layout, sensors, actuators, DAQ channels and software architecture	System block diagram and wiring plan
Sensor build	Construct moisture probe, divider, LM358 buffer and RC filter	Stable analog voltage output
Calibration	Measure dry, low, normal and wet soil samples	Voltage vs moisture curve and equation
LabVIEW dashboard	Create indicators, controls, graphs and logging	Working front panel and block diagram
Arduino control	Program motor control, pump timing and station detection	Robot moves and stops at pot markers
AI + website	Capture images, train/test or implement OpenCV method, create dashboard	Weed status table and alerts
Integration	Connect LabVIEW, DAQ, Arduino and Python results	End-to-end automatic pot scanning cycle
Evaluation	Run repeated sensor, robot, pump and AI tests	Performance results for final report

7.2 Performance evaluation plan

Area	Test	Parameter to report
Moisture sensor	Calibration using 4 soil conditions	Voltage vs moisture %, calibration curve, R^2 if fitted
Moisture sensor	Repeatability test	Same soil measured 5 times: mean, range, standard deviation
Moisture sensor	Response time	Time taken to stabilize after inserting probe
Temperature sensor	Comparison with thermometer	Error in °C
Pump control	Commanded vs actual pump duration	Timing error and maximum limit operation
Watering effect	Moisture before and after watering	Increase in moisture % after pump action
AI detection	Test image set classification	Accuracy, false positive rate, false negative rate
Robot navigation	Pot stopping test	Distance error from pot marker and successful stops/total trials
Full system	Cycle test over 3 pots	Total cycle time, logged values, successful decisions

7.3 Acceptance criteria

- The robot should stop at each pot station in at least 80% of trials during classroom demonstration.
- The moisture sensor should show a clear voltage difference between dry, normal and wet soil.
- LabVIEW should display calibrated moisture percentage and temperature without manual calculation.
- Pump activation should match the calculated duration within a reasonable timing error.
- The system should correctly log pot ID, moisture %, temperature, weed status and pump action.
- Weed detection should be reliable under the same camera angle and lighting used for training/demo.

8

Implementation Plan, Resources and Expected Outcomes

Practical work plan for completing the prototype

8.1 Hardware and software resources

Component	Quantity	Purpose
Existing robot chassis with DC motors	1	Mobile platform
Arduino Uno	1	Motor, pump and marker sensor control
L298N motor driver	1	Drive DC motors
NI DAQ card	1	Analog sensor acquisition and possible digital output
Laptop with LabVIEW	1	Main instrumentation dashboard and control logic
Webcam / phone / USB camera	1	Image input for weed detection
Mini water pump + tube + water container	1 set	Automatic irrigation
Relay module or MOSFET driver	1	Pump switching circuit
LM35 or DHT22	1	Temperature measurement
Stainless-steel probes	2	Self-built moisture sensor electrodes
LM358 op-amp	1	Voltage buffer for signal conditioning
Resistors and capacitors	Several	Divider and RC filter
IR sensor / ultrasonic sensor	1	Pot station detection
LEDs / buzzer	2-3 / 1	Status and alarm indication
Small plant pots	3-5	Test/demo environment

Software tools: LabVIEW, NI DAQmx or available DAQ interface, Arduino IDE, Python, OpenCV or a simple image classification tool, Flask for local web dashboard, and CSV/Excel for logging.

8.2 Suggested group division

Group member role	Main responsibility
Person A: Robot and pump control	Robot movement, station detection, motor driver, pump driver, Arduino code and actuator tests.
Person B: Sensors and DAQ	Moisture sensor build, signal conditioning, DAQ input, temperature sensor, calibration curves and LabVIEW measurement display.
Person C: AI, camera and website	Image capture, weed detection, Flask dashboard, CSV/JSON updates and connection between AI result and LabVIEW log.

8

Seven-Day Implementation Schedule and Demo Sequence

A realistic build order for a mini project

Day	Main activities	Output by end of day
1	Confirm final scope, pot layout, hardware list, wiring plan; test robot movement and pump separately.	Fixed project scope and basic hardware checked.
2	Build moisture probe, voltage divider, LM358 buffer and RC filter; read voltage with DAQ/Arduino.	Self-built moisture sensor gives measurable voltage.
3	Prepare dry/wet soil samples, record calibration readings, test temperature sensor, start LabVIEW display.	Moisture calibration table and temperature reading.
4	Add IR/ultrasonic station detection and test Pot 1 to Pot 3 movement; set up serial handshake.	Robot stops at pot stations and reports status.
5	Capture images, implement AI/OpenCV weed detection, create website dashboard and CSV/JSON update.	Website shows plant status and weed alert.
6	Integrate full cycle: stop, measure, detect, decide, water, log and move next.	End-to-end system demonstration.
7	Repeat testing, collect graphs, clean wiring, label components, prepare presentation and demo video.	Final presentation-ready prototype.

Final demonstration sequence

- Start LabVIEW and open the local website dashboard.
- Robot moves to Pot 1 and stops at the marker.
- DAQ reads soil moisture and temperature; LabVIEW displays raw and calibrated values.
- Camera captures the pot image and AI returns weed/no-weed result.
- LabVIEW calculates pump duration based on moisture deficit.
- Pump waters automatically if moisture is below target; otherwise no watering occurs.
- Website updates Pot 1 status and LabVIEW logs the data.
- Robot moves to Pot 2 and repeats the same cycle.

Expected outcomes

- A working prototype that scans 3-5 pot plants automatically.
- A calibrated self-built soil moisture sensor with a documented conditioning circuit.
- A LabVIEW dashboard showing sensor values, pump control, robot status and graphs.
- An Arduino-controlled mobile base and pump system.
- A local website showing weed alerts, moisture percentage and watering action.
- A data log that supports the final report and performance evaluation.

9

Risks, Limitations, Future Improvements and References

How to keep the project reliable and defendable

9.1 Main risks and mitigation

Risk	Effect	Mitigation
Moisture sensor readings drift due to corrosion or soil variation	Poor calibration reliability	Use stainless-steel probes, short-pulse excitation, repeat readings and average values.
Pump draws too much current	Arduino damage or reset	Use relay/MOSFET driver, flyback diode, external supply and common ground.
AI fails under new lighting	Wrong weed result	Use fixed camera angle and lighting; collect images under demo conditions.
Robot misses pot station	Measurement cycle starts at wrong place	Use black tape markers, slow speed near stations and repeated stop tests.
DAQ/Arduino communication delay	Timing mismatch	Use simple serial commands and acknowledgements such as POT_REACHED and PUMP_DONE.
Overwatering	Plant/pot flooding	Use maximum pump duration limit and test water volume before demo.

9.2 Limitations

The prototype is designed for a controlled indoor pot plant arrangement. It does not aim to operate as a full agricultural robot. The weed detection method is optimized for a fixed camera view and controlled lighting. Soil moisture calibration is relative unless gravimetric drying is performed. Watering control is based on pump duration rather than measured water flow.

9.3 Future improvements

- Add an LDR/light sensor and automatically control an LED grow light in timed intensity steps.
- Add a water flow sensor for more accurate irrigation volume feedback.
- Use a trained object detection model to locate weed position in the pot image.
- Add a servo marker or simple weed removal tool after detection.
- Add cloud storage or a mobile app after the local Flask dashboard is stable.
- Improve navigation using QR markers, RFID tags or line-following markers.

Final recommendation

Keep the project controlled and achievable. The strongest technical contribution is: calibrated self-built soil moisture sensor + LabVIEW/DAQ measurement + AI weed alert + automatic irrigation control.

9.4 References / source materials

[1]	University of Moratuwa, Sri Lanka. EE2043 / EE2044 Electrical Measurements and Instrumentation Mini Project Brief.
[2]	Project planning notes provided for: Smart Garden Robot for Pot Plant Monitoring, AI Weed Detection, and Automatic Irrigation Using LabVIEW-Based Instrumentation.
[3]	Arduino Uno, NI DAQ, LabVIEW, LM35, LM358 and OpenCV documentation should be cited in the final report after exact component/software versions are confirmed.

Suggested statement to lecturer

Project defense statement

We modified our original weed-control robot into a smart garden instrumentation robot. The system uses a self-built soil moisture sensor with signal conditioning and calibration, a temperature sensor, camera-based AI weed detection and navigation sensing. LabVIEW and the DAQ card acquire and display calibrated sensor data, while Arduino controls robot movement and irrigation. The system calculates pump duration from soil moisture percentage and updates weed alerts and plant conditions on a web dashboard.

Final submission item	Status to prepare before presentation
Circuit diagram	Show soil moisture divider, LM358 buffer, RC filter and pump driver
LabVIEW screenshots	Show front panel, DAQ channels, graphs, serial commands and data log
Calibration evidence	Include voltage readings for dry, low, normal and wet soil
Demo video/photos	Capture robot stopping, measuring, detecting weed and watering
Performance table	Report sensor repeatability, pump timing, AI accuracy and stopping success